

Graph Attention Transformer With Multitask Learning for Motion Prediction in Autonomous Driving

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Abstract—The motion prediction of surrounding vehicles is crucial for autonomous driving. As it can effectively guide the planning and decision-making of the autonomous vehicle. Although many methods have been proposed to predict the future motion of vehicles, but they failed capture the essence of multitask learning for vehicle motion prediction and lacked effective semantic and interaction information fusion. In this article, we propose a novel model, graph attention transformer with multitask learning, to predict the future motion of the target vehicle by modeling motion prediction task as a multitask learning problem of trajectory prediction and driving intention estimation. The model consists of three modules: feature fusion, spatio-temporal interaction, and motion decoder. The feature fusion module is used to effectively fuse the semantic and dynamic information of the target vehicle. The Spatial Interaction module is used to model the interaction between vehicles. The motion decoder module decouples the motion prediction into multitask learning for trajectory prediction and driving intention estimation in a parallel manner. Extensive experiments on real-world datasets, with average improvement of 50% in NGSIM and 40% in HighD, demonstrate that the proposed model achieves superior performance.

Impact Statement—Motion prediction is a popular research topic in autonomous driving as it significantly contribute to safety and efficiency of autonomous driving. However, the existing methods failed to capture the essence of multitask learning for vehicle motion prediction and lacked effective semantic and interaction information fusion. Thus resulting in unsatisfactory performance especially in long-term prediction and strong interaction scenarios. The GATran model we introduced in this article overcomes these limitations. With a significant improvement of average 50% in NGSIM and 20% in HighD compared to the state-of-the-art methods, the proposed model demonstrates superior performance

in motion prediction tasks. Thus showing the potential of the proposed model to be applied in the autonomous driving system for better safety and efficiency in the real-world scenarios.

Index Terms—Autonomous driving, graph attention transformer (GATran), mixture of experts, motion prediction, multitask learning.

I. INTRODUCTION

THE motion behavior of surrounding vehicles exerts a significant influence on the driving behavior of the subject vehicle, underscoring the critical importance of motion prediction in the context of autonomous driving. The objective of motion prediction is to forecast the future trajectory of the target vehicle by leveraging its historical movement data along with information regarding the motion of nearby vehicles. This task is inherently challenging due to the complexity and dynamism of the traffic environment, which includes uncertainties in the motion patterns of surrounding vehicles, the variability of individual driving behaviors, and the intricate interactions between vehicles.

Traditional motion prediction methods primarily rely on manually defined physics-based models, such as social force models [1]. However, these approaches often struggle to accommodate the complexity and variability of dynamic traffic environments. The emergence of data-driven deep learning techniques has significantly advanced vehicle motion prediction, leading to numerous breakthroughs. Early methods based on CNNs [2] and recurrent neural networks (RNNs), including social LSTM [3], convolutional social pooling [4], and social GAN [5], demonstrated the ability to effectively capture both spatial and temporal information of the target and surrounding vehicles while modeling the temporal dependencies of historical trajectories. Despite their contributions, these approaches fail to fully leverage the principles of multitask learning in vehicle motion prediction and lack robust mechanisms for semantic and interaction information fusion.

Recently, graph neural networks (GNNs) and transformers have garnered significant attention from researchers due to their remarkable success in various individual tasks. Transformers [6], [7] are good at capturing temporal information, while graph networks [8], [9], [10], [11] are good at aggregating spatial information with larger receptive fields. Some researchers [12],

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[13], [14], [15], [16] have proposed trajectory prediction methods that combine transformer and graph networks to simultaneously capture and aggregate spatiotemporal information, achieving excellent performance in solving various problems. Nevertheless, existing methods still struggle to effectively integrate rich semantic and dynamic features, capture complex spatio-temporal interactions among agents, and leverage auxiliary tasks through multitask learning, which together limits their overall prediction performance in complex traffic scenarios. To overcome these challenges, we propose a novel framework, graph attention transformer with multitask learning (GATran). This model introduces an effective module for fusing the semantic and dynamic information of the target vehicle, while integrating a graph attention mechanism with transformers to capture spatio-temporal interactions between vehicles. Unlike existing GNN-transformer approaches [17], [18], [19] that perform interaction modeling on temporally aggregated agent representations, our framework embeds graph attention directly into the transformer encoder and constructs frame-level interaction graphs across all observed time steps, preserving fine-grained temporal dynamics. Furthermore, the motion prediction task is reformulated as a multitask learning problem, encompassing trajectory prediction and driving intention estimation. To achieve this, we incorporate a multigate mixture-of-experts (MMoE) module into the transformer decoder, enabling a more robust and comprehensive modeling approach. The main contributions of this article can be summarized as follows.

- 1) *Feature Fusion Module*: We propose a feature fusion mechanism that effectively embeds heterogeneous data, including semantic lane information and dynamic agent states, into a unified latent space, providing a robust foundation for subsequent interaction modeling.
- 2) *Spatio-Temporal Interaction Module*: We design a novel spatio-temporal interaction module that integrates graph attention networks (GAT) directly into the transformer architecture. Unlike standard methods that aggregate history beforehand, our approach captures frame-level spatial interactions and temporal dependencies explicitly, preserving fine-grained motion dynamics.
- 3) *Multitask Learning Framework*: We integrate the multigate mixture-of-experts (MMoE) directly into the transformer decoder layers to explicitly disentangle representations, allowing the model to learn robust task-specific features for trajectory forecasting and intention estimation from shared spatio-temporal patterns.

II. RELATED WORK

A. Multiagent Interaction

Multiagent interaction between vehicles is a critical aspect of vehicle trajectory prediction [20]. Traditional approaches to multiagent interaction modeling rely on manually defined physics-based models, such as the social force model [1]. In this framework, interactions between vehicles are described by defining attractive and repulsive forces. While these

methods provide a theoretical basis for explaining certain interactive behaviors, they struggle to account for the complexity and dynamic nature of real-world traffic environments.

The emergence of data-driven deep learning techniques has significantly advanced vehicle trajectory prediction, leading to notable progress in multiagent interaction modeling. Early methods introduced the pooling mechanism, borrowed from traditional convolutional neural networks (CNNs), into interaction modeling. These approaches [3], [4], [21] incorporate a social pooling layer to process interaction information based on encoder features. Attention mechanisms, inspired by the way traffic participants process information, have also been effectively employed to enhance social interaction modeling. Some methods [22], [23] utilize attention weights to represent the degree of influence between vehicles, improving the accuracy of interaction representations. Furthermore, traffic scenes can be naturally represented as graph structures, where each traffic participant corresponds to a node, and their interactions are captured by the edges connecting these nodes [24]. The development of GNNs has facilitated their application in interaction modeling. Graph convolutional networks (GCNs), an extension of convolutional operations to graph structures, have been employed in methods such as [25] and [26] to extract spatio-temporal interaction features and model multiagent interactions from both spatial and temporal perspectives. Graph attention networks (GATs) further enhance message passing between nodes. For instance, methods like social BiGAT [17] and STGAT [18] leverage GATs to manage hidden variables in LSTM models, thereby extracting richer interaction information between agents. Graph-based methods have shown great potential in capturing the complex and dynamic interactions between vehicles.

In addition, the interaction in multiagent scenarios is not only between vehicles, but also more complex interactions such as between vehicles and the environment, maps, and interactions between different traffic participants. Some methods utilize semantic information for modeling spatio-temporal interactions [3], [25], [27], [28]. As an auxiliary information for trajectory prediction, semantic information does not directly generate predicted trajectories but is combined with trajectory prediction modeling methods to produce more accurate prediction information [11], [16]. Currently, some studies have considered the impact of semantic information on trajectory prediction [29], [30], [31], with the majority focusing on semantic information such as object positions, road structures, and categories of traffic participants in the scene. Sophie [32] proposed by Sadeghian et al. uses CNN to process scene semantic information; LaneGCN [8] directly constructs lane maps from raw map data to capture complex topological structures and long-term dependencies of lane maps; semantics-STGCNN [29] proposed by Rainbow et al. adds categories of traffic participants to the GCN to better predict trajectories; MULTICASS-SGCN [30] proposed by Li et al. is a multiclass trajectory prediction method based on sparse GCN, which considers proxy label information and uses a new interaction mask to adaptively determine the spatial and temporal connections of proxies based on their interaction scores. Of course, current methods for modeling interactions

in multiagent scenarios remain limited, as spatio-temporal interactions are often underexplored and the fusion of semantic and dynamic features remains insufficient for comprehensive interaction reasoning. In our work, we propose a new model that combines the graph attention mechanism with Transformer to capture spatio-temporal interactions between vehicles, thus improving the accuracy of modeling multiagent interactions.

B. Multimodal Distribution Prediction

Due to variations in individual driving styles and road conditions, vehicle driving behavior exhibits significant diversity [33], [34]. Traditional unimodal prediction approaches are unable to capture this variability, as they provide a single deterministic trajectory prediction. Consequently, recent research has shifted toward multimodal trajectory prediction, which aims to generate diverse future trajectory distributions based on a single ground truth [27]. These approaches can be broadly categorized into two methodological types. The first category involves introducing random noise into the model to represent the diversity of driving behavior through latent variables. Generative deep learning models, such as generative adversarial networks (GANs) and conditional variational autoencoders (CVAEs), are commonly employed in this approach. These models sample latent variables from random noise, which are then fused with encoded features to generate diverse trajectory predictions. Gupta et al. [5] were the first to employ GANs for multimodal trajectory prediction, combining the encoded information of historical trajectories with Gaussian noise vectors. Similarly, CVAE-based models [35] offer greater control and refinement over the distribution of latent variables, enabling improved modeling of multimodal behaviors. However, this approach has certain limitations: it often lacks interpretability, and random sampling may fail to produce trajectories that adequately cover all possible scenarios. The second category defines explicit conditions within the model to capture driving diversity, such as driving intentions or anchor points. For instance, Hasan et al. [36] employed LSTMs to predict multiple potential driving intentions along with their associated probabilities. Other methods [10], [11], [37], associate trajectory predictions with specific targets, generating multimodal trajectories oriented toward those targets. Furthermore, methods such as that proposed in [38] adapt multimodal trajectories to road layouts, producing variable numbers of driving intentions based on diverse road geometries. This approach provides a more interpretable framework for multimodal trajectory prediction, as it leverages explicit conditions to better explain the variability of driving behaviors. However, current methods may lack certain interpretability or struggle to effectiveness of multimodal trajectory generation. In our work, we integrate the learnable query mechanism with the Gaussian distribution to predict the multimodal trajectory distribution in a more interpretable and effective manner. The Gaussian distribution, characterized by its mean (μ) and standard deviation (σ), is a symmetric, bell-shaped probability distribution that provides a principled framework for modeling uncertainty in vehicle trajectory prediction. In this domain, future positions of a vehicle are commonly

modeled as random variables drawn from a Gaussian distribution, where the mean represents the most likely location and the variance (or covariance) captures prediction uncertainty. This probabilistic formulation enables models to move beyond deterministic outputs and better reflect the inherently stochastic nature of human driving behavior. Crucially, in multimodal trajectory prediction—where multiple plausible future paths may exist due to complex traffic dynamics or ambiguous driver intentions—the Gaussian distribution is often employed within mixture density models or integrated into generative frameworks to capture diverse behavioral modes. During training, the Gaussian likelihood allows for the use of negative log-likelihood (NLL) loss functions, promoting simultaneous learning of both accuracy and confidence. Furthermore, treating the distribution parameters (mean and variance) as learnable allows models to dynamically adapt to varying traffic contexts and input features. The mathematical tractability, differentiability, and interpretability of the Gaussian distribution make it particularly well-suited for integration into deep learning architectures, where it plays a central role in producing reliable, uncertainty-aware, and multimodally expressive trajectory forecasts.

C. Graph Neural Networks for Trajectory Prediction

While CNNs effectively capture the characteristics of the local scene, their limited receptive fields hinder their ability to fully exploit the structured nature of driving environments. Recent advances address this by integrating vectorized map representations with GNNs to model spatial and semantic relationships in a more structured and efficient manner. LaneGCN [8] constructs a lane graph where each segment is represented as a node. Graph convolutions propagate structural information, while an attention-based FusionNet fuses lane and trajectory features, capturing complex topologies and long-range dependencies. GOHOME [9] builds upon HOME by using lane graphs and cross-attention to update agent features with map information. It predicts the probability distribution over lanes, selects top candidates, and generates heatmaps to guide goal-conditioned trajectory prediction. Compared to HOME, it significantly reduces memory usage and computation. TNT [10] models multimodal uncertainty by predicting endpoint anchors using a VectorNet-style encoder. These anchors serve as priors for a decoder that regresses complete trajectories. A ranking module selects the top- y most likely outcomes. DenseTNT [11] removes anchor dependency by scoring lane pixels with attention-based mechanisms, enabling dense endpoint sampling and more flexible multimodal predictions. LaneRCNN [31] introduces LaneRoI, a representation that encodes both lane and agent features via a graph encoder. It incorporates agent-agent interactions and regresses full trajectories from predicted endpoints, improving precision and reducing implausible outputs.

To address limitations of transformer architectures with structured data, STAR [12] proposes TGConv, a transformer-based graph convolution that performs message passing via self-attention and captures agent interactions and temporal dynamics. It achieves strong performance on standard benchmarks. LAformer [13] combines VectorNet with attention

mechanisms to filter out irrelevant lanes. A Laplace mixture density network decoder estimates both position and scale, excelling in ballistic prediction tasks. Hivt [14] adopts a hierarchical framework separating local (agent-centric) and global (interagent) interactions. Local GNNs extract nearby dynamics, while attention models broader context, offering a balance between accuracy and efficiency. QCNet [15] enhances rotation and translation invariance by encoding trajectories relative to fixed reference points. Graph-based encoders and attention-based decoders refine anchor trajectories, improving generalization and reducing complexity. QCNeXt [16] extends QCNet by modeling temporal interactions between agents using cross-attention, self-attention, and column-wise modules. It demonstrates that multiagent models can outperform single-agent counterparts, even in single-agent settings.

III. METHODOLOGY

A. Problem Formulation

We formulate the traffic scenario as a set of vehicles $\mathcal{V} = \{V_0, V_2, \dots, V_N\}$, where V_0 (we uniformly use subscript 0 to represent the target vehicle) is denoted as the target vehicle and V_i is denoted as the i th vehicle in the scene. Each vehicle V_i has a semantic class (i.e., car, truck, or bus) and is represented by a sequence of states $X_i = \{s_i^{-t_h}, s_i^{-t_h+1}, \dots, s_i^0\}$, where s_i^t is the state of the i th vehicle at time t . The state of each vehicle s_i^t is defined as $s_i^t = \{x_i^t, y_i^t, v_i^t, a_i^t, l_i^t\}$, where x_i^t and y_i^t are the coordinates of the i th vehicle at time t , v_i^t is the velocity of the i th vehicle at time t , a_i^t is the acceleration of the i th vehicle at time t , and l_i^t is the lane of the i th vehicle at time t . Assuming the current time is $t = 0$, given the historical states of all vehicles for the past t_h time steps $X_{0:N} = s_{0:N}^{-t_h:0}$, where X_0 represent the historical states of target vehicle, $X_{1:N}$ represent historical states of neighbor vehicles, the goal is to predict K possible future trajectories of the target vehicle V_0 for the next t_f time steps $Y = \{Y_k | k = 1, \dots, K\}$, $Y_k = \hat{y}^{1:t_f}$, where $\hat{y}^t$ is the predicted trajectory of the target vehicle at time t , and its corresponding driving intention $I = \{I_k | k = 1, \dots, K\}$, $I_k = \{I_{\text{lat}}, I_{\text{lon}}\}$, where $I_{\text{lat}} \in (\mathcal{C}\mathcal{L}\mathcal{L}, \mathcal{L}\mathcal{K}, \mathcal{C}\mathcal{L}\mathcal{R})$ and $I_{\text{lon}} \in (\mathcal{A}\mathcal{C}\mathcal{C}, \mathcal{C}\mathcal{O}\mathcal{N}, \mathcal{D}\mathcal{E}\mathcal{C})$ are the lateral and longitudinal driving intentions of the target vehicle, respectively. Additionally, $\mathcal{C}\mathcal{L}\mathcal{L}, \mathcal{L}\mathcal{K}, \mathcal{C}\mathcal{L}\mathcal{R}$ represent the left lane change, keep lane, and right lane change, respectively, and $\mathcal{A}\mathcal{C}\mathcal{C}, \mathcal{C}\mathcal{O}\mathcal{N}, \mathcal{D}\mathcal{E}\mathcal{C}$ represent the acceleration, constant speed, and deceleration, respectively.

B. Model Architecture

Fig. 1 illustrates the architecture of the proposed model, GATran, which is composed of three key modules: 1) a module integrates rich semantic and dynamic information with the historical trajectory of the target vehicle, enabling a comprehensive feature representation; 2) a component models the interactions between the target vehicle and its surrounding vehicles, capturing spatio-temporal dependencies critical for accurate motion prediction; and 3) a module predicts both the future trajectory and driving intention of the target vehicle, leveraging a multitask learning framework to enhance prediction performance.

The detailed design and functionality of each module are described in the following sections.

C. Feature Fusion Module

Current feature extraction methods often focus on the historical trajectory of the target vehicle, neglecting the semantic and dynamic information that could significantly enhance the feature representation. In order to extract the more abundant feature representation of the target vehicle, we introduce a feature fusion module that incorporates the semantic and dynamic information of the target vehicle as is shown in Fig. 2. In this module, we first extract and the embedding of historical trajectory of the target vehicle using a MLP layer into a latent space as follows:

$$Q = \mathcal{MLP}_q(X_0\{ : 2 \}) \quad (1)$$

where $X_0\{ : 2 \}$ is the historical trajectory of the target vehicle, and Q is the embedding of the historical trajectory. Then, we extract the semantic and dynamic information of the target vehicle and embedding them into a latent space as follows:

$$K = \mathcal{MLP}_k(X_0\{3 : 5\}) \quad (2)$$

$$V = \mathcal{MLP}_v(X_0\{3 : 5\}) \quad (3)$$

where $X_0\{3 : 5\}$ is the semantic and dynamic information of the target vehicle, which contains the vehicle velocity, acceleration, and lane, and K and V are the embeddings of the semantic and dynamic information.

Subsequently, we aggregate the embeddings of the historical trajectory and the semantic and dynamic information of the target vehicle using a temporal encoding module with a transformer as follows:

$$\begin{cases} \hat{Q} = K = V = \mathcal{PE}(Q) \\ H = \mathcal{MSATTN}(\hat{Q}, K, V) \\ \hat{H} = \varphi(H + Q) \end{cases} \quad (4)$$

where $\mathcal{PE}$ is the positional encoding, $\mathcal{MSATTN}$ is the multihead self-attention, and φ is the residual connection. Finally, we obtain the feature representation of the target vehicle by concatenating the encoding information $\hat{H}$ and embedding information of semantic and dynamic information K and V by a multihead attention mechanism as follows:

$$\begin{cases} Q = \hat{H} \\ \hat{F} = \mathcal{MATTN}(Q, K, V) \\ F = \varphi(\hat{F} + Q) \end{cases} \quad (5)$$

where $\mathcal{MATTN}$ is the multihead attention mechanism, and F is the feature representation of the target vehicle.

D. Spatio-Temporal Interaction Module

Graph structure could effectively represent the spatial relationship between vehicles in the traffic scene. However, the existing methods often focus on the spatial interaction modeling of graph structure, neglecting the temporal dependencies of historical states. To address this limitation, we propose a

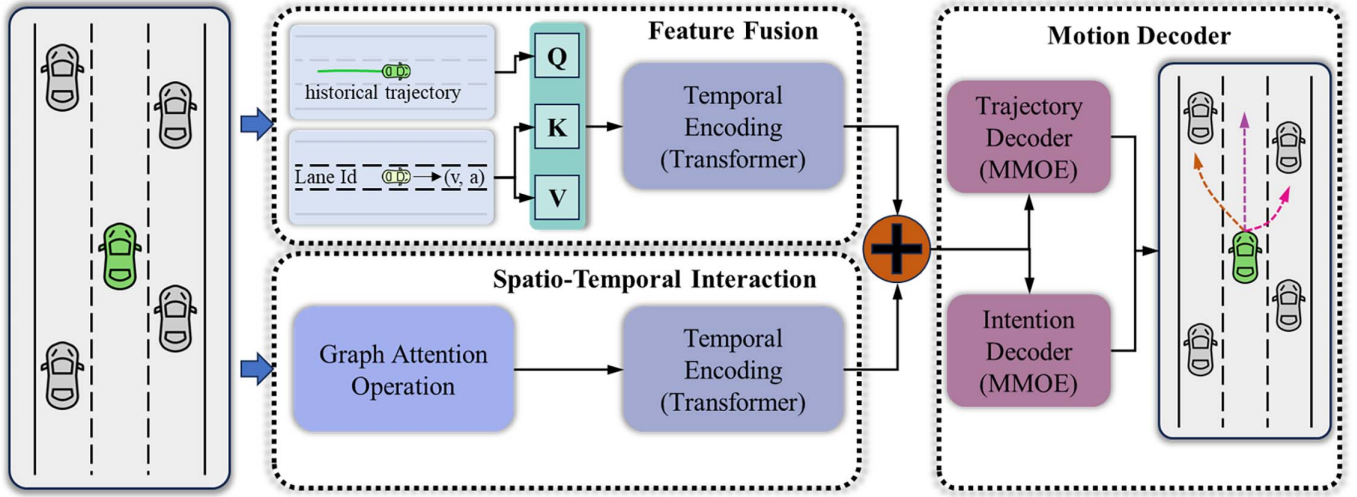


Fig. 1. Architecture of the proposed model. The model consists of three modules: feature fusion, spatio-temporal interaction, and MMOE. The feature fusion module is used to extract the feature representation of each vehicle. The spatio-temporal interaction module is used to model the interaction between vehicles. The MMOE module is used to predict the future trajectory of each vehicle.

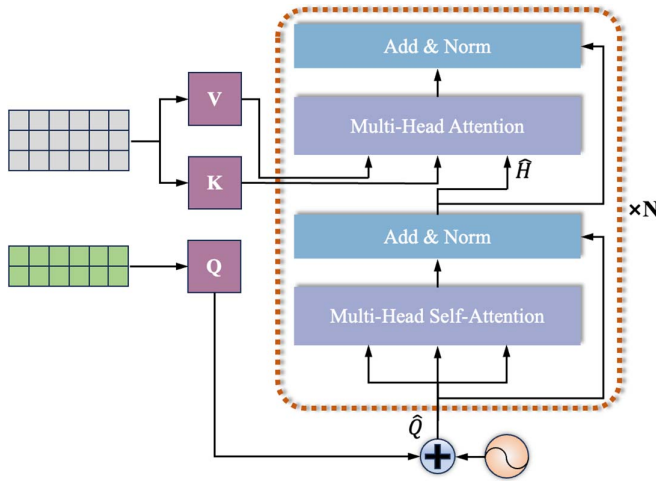


Fig. 2. Architecture of the feature fusion module. The module extracts the feature representation of the target vehicle by incorporating the semantic and dynamic information of the target vehicle.

spatio-temporal interaction module that captures the interaction between vehicles as is shown in Fig. 3 through a graph attention mechanism with transformers. Formally, we first construct a graph structure $\mathcal{G}_t = (\mathcal{H}_t, \mathcal{E}_t)$ at each time step t , where $\mathcal{H}_t = h_t^0, h_t^1, \dots, h_t^N$ is the set of nodes representing encoding embeddings of vehicles and $\mathcal{E}_t = e_t^{ij}$ is the set of attention weights between nodes representing the interaction between vehicles. Then, we perform a graph attention mechanism to model the interaction between vehicles as follows:

$$e_t^{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^T \left[\mathbf{W}h_t^i \parallel \mathbf{W}h_t^j\right]\right)\right)}{\sum_{k \in \mathcal{N}_i} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^T \left[\mathbf{W}h_t^i \parallel \mathbf{W}h_t^k\right]\right)\right)} \quad (6)$$

where $\mathbf{W}$ and $\mathbf{a}$ are the weight matrix, $\mathcal{N}_i$ is the set of neighbors of node i , and $\parallel$ is the concatenation operation. After that,

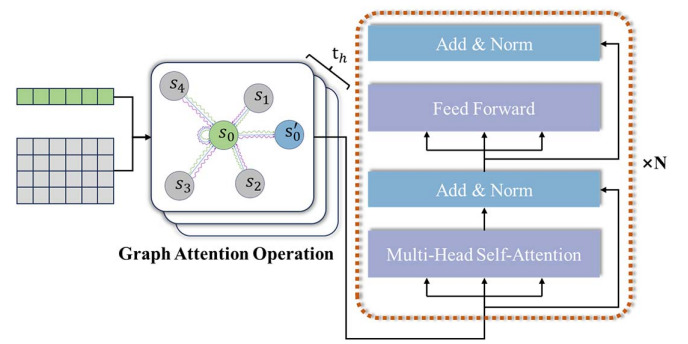


Fig. 3. Architecture of the spatio-temporal interaction module. The module models the interaction between vehicles by incorporating the graph attention mechanism with transformer.

we aggregate the interaction information to obtain the updated feature representation of node i as follows:

$$\hat{h}_t^i = \sigma \left(\sum_{j \in \mathcal{N}_i} e_t^{ij} \mathbf{W}h_t^j \right) \quad (7)$$

where σ is the activation function, e_t^{ij} is the attention weight between node i and node j , and $\hat{h}_t^i$ is feature representation of node.

Through the aforementioned process, we obtain the spatial interaction features at each timestep. However, trajectory prediction not only relies on interactions in the spatial domain but also heavily depends on temporal correlations across different timesteps. To effectively capture these cross-temporal dependencies, we further introduce a temporal encoding module based on self-attention mechanism to explicitly model temporal interaction features

$$\begin{cases} h_{st}^i = \text{MSATTN}(\hat{h}_{-t_h:0}^i, \hat{h}_{-t_h:0}^i, \hat{h}_{-t_h:0}^i) \\ \hat{h}_{st}^i = \varphi(h_{st}^i + \hat{h}_{-t_h:0}^i) \end{cases} \quad (8)$$

here, $SATTN$ represents the multihead self-attention mechanism, φ is the residual connection and h_{st}^i denotes the aggregated spatio-temporal interaction features of node i , encapsulating both spatial information from neighboring nodes and temporal dependencies across multiple timesteps.

Finally, a feed-forward network is employed to further refine and extract features through nonlinear transformations, enabling the capture of higher-level and more discriminative representations, thereby effectively improving the overall performance of the trajectory prediction task

$$\begin{cases} h_o^i = \mathcal{MLP}(\hat{h}_{st}^i) \\ \hat{h}_o^i = \varphi(h_o^i + \hat{h}_{st}^i) \end{cases} \quad (9)$$

here we utilize $\mathcal{MLP}$ as feed forward network and φ denotes the residual connection.

E. Motion Decoder Module

Motion prediction refer to different tasks with different characteristics while the existing methods often treat them as a single trajectory prediction task, which may lead to suboptimal performance. Different from previous multimodal trajectory prediction methods, we decouple the motion prediction into multitask learning for multimodal trajectory prediction and driving intention estimation. To improve the performance of multitask learning, we propose a motion decoder module that integrates transformer with multigate mixture-of-experts (MMoE) as is shown in Fig. 4. We replace the feed-forward network in the transformer with MMoE to learning the multitask latent information by sharing the parameters of the experts and the gating network. The MMoE consists of multiple experts and a gating network, where each expert is responsible for learning a specific task, and the gating network is used to determine the importance of each expert for each task. The MMoE can effectively capture the complex and diverse distribution of multimodal trajectory prediction and driving intention estimation. The MMoE is defined as follows:

$$\begin{cases} f^k(\tilde{h}) = \sum_{i=1}^n g^k(\tilde{h})_i f_i(\tilde{h}) \\ g^k(\tilde{h}) = \text{Softmax}(W_{gk}\tilde{h}) \end{cases} \quad (10)$$

where $\tilde{h}$ is the output of standard structure of transformer, f^k is the output of the k th subtask, g^k is the gating network that combines the results of experts in the k th subtask. Each task has an independent gating network.

In the end, we use two different types of prediction heads to predict the future trajectory and driving intention of the target vehicle. The prediction head for trajectory prediction is a linear layer that predicts the future trajectory of the target vehicle, and the prediction head for driving intention estimation is a softmax layer that predicts the driving intention

$$\begin{cases} y = \mathcal{MLP}_{\text{traj}}(\xi(e)) \\ I = \text{softmax}(\mathcal{MLP}_{\text{int}}(\xi(e))) \end{cases} \quad (11)$$

where $\mathcal{MLP}_{\text{traj}}$ and $\mathcal{MLP}_{\text{int}}$ are the linear layer, ξ denoted the $Add\&Norm$ operation, e is the output of the MMoE, y is the

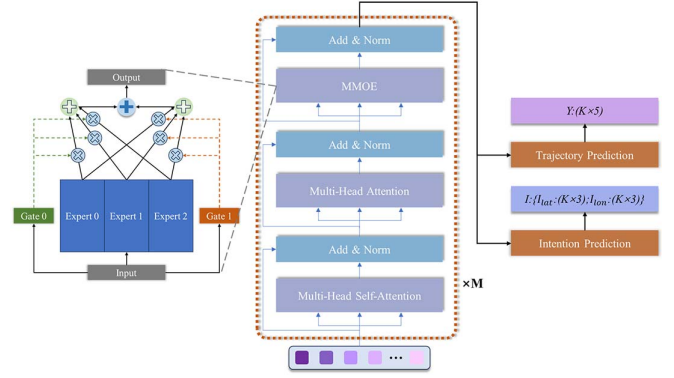


Fig. 4. Architecture of the motion decoder module. The module integrates transformer with multigate mixture-of-experts (MMoE) to predict the future trajectory and driving intention in a multitask learning manner.

predicted trajectory of the target vehicle, and I is the predicted driving intention of the target vehicle.

IV. EXPERIMENTS

A. Training and Implementation Details

We train the model through a one stage training process, where the model is trained end-to-end. For optimizing the multitask learning framework, we set the different loss function each task. Specifically, the trajectory prediction task is optimized using negative log-likelihood loss as a regression task, and the driving intention estimation task is optimized using cross-entropy loss as a classification task. The overall loss function is defined as follows:

$$\mathcal{L} = \mathcal{L}_t + \mathcal{L}_i \quad (12)$$

where $\mathcal{L}_t$ and $\mathcal{L}_i$ are the loss functions for trajectory prediction and driving intention estimation, respectively. The $\mathcal{L}_t$ is defined as follows:

$$\mathcal{L}_t = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^{t_f} \log(\hat{y}_i^t, y_i^t) \quad (13)$$

where N is the number of samples, t_f is the prediction horizon, $\hat{y}_i^t$ is the ground truth trajectory of the i_{th} sample at time t , and y_i^t is the predicted trajectory of the i_{th} sample at time t . The $\mathcal{L}_i$ is defined as follows:

$$\mathcal{L}_i = \mathcal{L}_{\text{lat}} + \mathcal{L}_{\text{lon}} \quad (14)$$

$$\mathcal{L}_{\text{lat}} = \mathcal{L}_{\text{cro}}(\hat{I}_{\text{lat}}, I_{\text{lat}}) \quad (15)$$

$$\mathcal{L}_{\text{lon}} = \mathcal{L}_{\text{cro}}(\hat{I}_{\text{lon}}, I_{\text{lon}}) \quad (16)$$

where $\mathcal{L}_{\text{lat}}$ and $\mathcal{L}_{\text{lon}}$ are the loss functions for lateral and longitudinal driving intention estimation, respectively, I_{lat} and I_{lon} are the predicted driving intention of the target vehicle, $\hat{I}_{\text{lat}}$ and $\hat{I}_{\text{lon}}$ are the ground truth driving intention, and $\mathcal{L}_{\text{cro}}$ is the cross-entropy loss function.

For implementation details, in the feature fusion module, a single MLP with 64-dimensional outputs is used to generate query, key, and value vectors and the temporal encoding module stacks two layers ($N = 2$). In the spatio-temporal interaction

module, the GAT network adopts a 4-head attention mechanism, and the temporal encoding module also stacks two layers ($N = 2$). For the motion decoder module, we utilize three stacked decoder layers ($M = 3$), and the MMoE network is configured with three experts. Additionally, we initialize nine queries with a feature dimension of 128 to learn different motion modes. All other attention mechanisms consistently adopt an eight-head configuration.

The model is developed using PyTorch and trained on four NVIDIA RTX A6000 GPUs. We trained the model for 20 epochs with a batch size of 64 and Adam optimizer with an adaptive learning rate. The DDP distributed training strategy is used to accelerate the training process.

B. Datasets and Data Processing

1) *Datasets*: We conducted experiments using three real-world datasets: US-101, I-80, and HighD [49]. The US-101 and I-80 datasets are derived from the NGSIM dataset [50], which provides detailed vehicle state information for highways US-101 and I-80. The NGSIM dataset, recorded at a sampling frequency of 10 Hz, captures various traffic scenarios ranging from low to high density. The HighD dataset, on the other hand, was collected using drones over six distinct locations on German highways. It comprises trajectory data for over 11 050 vehicles, including detailed attributes such as vehicle type, size, and maneuvers, extracted using state-of-the-art trajectory extraction algorithms.

2) *Data Processing*: For the trajectory prediction task, we extract fixed-length trajectory segments of 8 s, where the first 3 s are used as historical observations and the subsequent 5 s serve as future motion targets. For the driving intention prediction task, ground-truth labels for both lateral and longitudinal maneuvers are constructed based on future vehicle behavior. Lateral driving intentions are categorized into three classes: lane keep (LK), left lane change (CLL), and right lane change (CLR). To capture the complete maneuver process rather than only the instantaneous lane-crossing moment, an asymmetric temporal window consisting of a 3 s historical horizon and a 5 s future horizon is employed. A frame is labeled as a lane change (CLL or CLR) if the lane ID at $t + 5s$ differs from that at t , or if the lane ID at t differs from that at $t - 3s$, thereby including both the preparation and stabilization phases of the maneuver. The lane-change direction is determined by the sign of the lane ID change. If the lane ID remains unchanged across $t - 3s$, t , and $t + 5s$, the frame is labeled as lane keep. Longitudinal driving intentions are classified into constant speed (CON), deceleration (DEC), and accelerating (ACC). Instead of relying on instantaneous acceleration thresholds, which can be sensitive to noise, a ratio-based strategy is adopted using average longitudinal speeds computed over asymmetric temporal windows. Specifically, the average longitudinal velocity is calculated over a 3 s historical window and a 5 s future window. Let v_{hist} and v_{fut} denote the historical and future average velocities, respectively, and define the ratio $r = v_{\text{fut}}/v_{\text{hist}}$. Frames are labeled as deceleration if $r < 0.8$, accelerating if $r > 1.25$, and constant speed

otherwise. Finally, the datasets were partitioned into training, validation, and test sets with a ratio of 7:1:2.

C. Metrics

We evaluate the performance of the proposed model using three widely adopted metrics: minimum root mean square error (minRMSE), minimum average displacement error (minADE), and minimum final displacement error (minFDE). minRMSE quantifies the average Euclidean distance between the predicted trajectory and the ground truth trajectory over the entire prediction horizon. minADE calculates the mean Euclidean distance between the predicted and ground truth trajectories at each time step throughout the prediction horizon. minFDE evaluates the Euclidean distance between the predicted trajectory and the ground truth trajectory specifically at the final time step. The mathematical formulations for minRMSE, minADE, and minFDE are as follows:

$$\text{minRMSE}_k = \sqrt{\frac{1}{B} \sum_{i=1}^B \min_k \left(\frac{1}{t_f} \sum_{t=1}^{t_f} \|\hat{y}_{i,k}^t - y_i^t\|^2 \right)} \quad (17)$$

$$\text{minADE}_k = \frac{1}{B} \sum_{i=1}^B \min_k \left(\frac{1}{t_f} \sum_{t=1}^{t_f} \|\hat{y}_{i,k}^t - y_i^t\|_2 \right) \quad (18)$$

$$\text{minFDE}_k = \frac{1}{B} \sum_{i=1}^B \min_k \|\hat{y}_{i,k}^{t_f} - y_i^{t_f}\|_2 \quad (19)$$

where B is the number of samples, t_f is the prediction horizon, k is the number of predicted trajectories, $\hat{y}_i^t$ is the predicted trajectory of the i_{th} sample at time t , and y_i^t is the ground truth trajectory of the i_{th} sample at time t .

D. Comparison Baselines

- 1) *S-LSTM* [3]: This model utilizes individual LSTMs for each trajectory and connects them via a social pooling layer to share hidden states, capturing interactions for joint trajectory prediction.
- 2) *CS-LSTM* [4]: This model uses a shared-weight LSTM encoder to learn vehicle dynamics and convolutional social pooling layers to understand spatial dependencies between vehicles.
- 3) *S-GAN* [5]: This model combines an encoder-decoder structure and an encoder discriminator to aggregate multi-agent information using a novel pooling mechanism and generate diverse trajectories.
- 4) *MATF-GAN* [39]: This model encodes scene context and the past trajectories of multiple agents, uses convolutional layers to capture interactions between agents, and employs an adversarial training mechanism decoder to predict future trajectories.
- 5) *WSiP* [40]: This model uses a novel wave superposition inspired pooling method within an encoder-decoder framework to dynamically aggregate high-order interactions among vehicles for improved trajectory prediction.

TABLE I
QUANTITATIVE RESULTS ON THE NGSIM AND HIGHD DATASETS

Dataset	Baselines	Prediction Horizons					
		1s	2s	3s	4s	5s	Ave
NGSIM	S-LSTM [3]	0.65 / 0.21 / 0.38	1.31 / 0.54 / 0.90	2.16 / 0.72 / 1.52	3.25 / 1.02 / 2.30	4.55 / 1.38 / 3.27	2.38 / 0.76 / 1.67
	CS-LSTM [4]	0.61 / 0.20 / 0.37	1.27 / 0.44 / 0.88	2.09 / 0.70 / 1.48	3.10 / 1.00 / 2.23	4.37 / 1.34 / 3.18	2.29 / 0.74 / 1.63
	S-GAN [5]	0.57 / - / -	1.32 / - / -	2.22 / - / -	3.26 / - / -	4.40 / - / -	2.35 / - / -
	MATF-GAN [39]	0.66 / - / -	1.34 / - / -	2.08 / - / -	2.97 / - / -	4.13 / - / -	2.22 / - / -
	WSiP [40]	0.56 / - / -	1.23 / - / -	2.05 / - / -	3.08 / - / -	4.34 / - / -	2.25 / - / -
	MHA-LSTM [41]	0.41 / - / -	1.01 / - / -	1.74 / - / -	2.67 / - / -	3.83 / - / -	1.93 / - / -
	FHIF [42]	0.40 / - / -	0.98 / - / -	1.66 / - / -	2.52 / - / -	3.63 / - / -	1.84 / - / -
	DACR-AMTP [43]	0.57 / - / -	1.07 / - / -	1.68 / - / -	2.53 / - / -	3.40 / - / -	1.85 / - / -
	HMNet [44]	0.50 / 0.16 / 0.31	1.13 / 0.38 / 0.79	1.89 / 0.62 / 1.36	2.85 / 1.91 / 2.08	4.04 / 1.23 / 2.95	2.08 / 0.66 / 1.50
	STDAN [45]	0.42 / 0.13 / 0.26	1.01 / 0.33 / 0.72	1.69 / 0.56 / 1.23	2.56 / 0.82 / 1.87	3.67 / 1.13 / 2.71	1.87 / 0.59 / 1.36
	iNATran [46]	0.39 / 0.12 / 0.25	0.96 / 0.32 / 0.69	1.61 / 0.54 / 1.19	2.42 / 0.79 / 1.80	3.43 / 1.08 / 2.55	1.76 / 0.57 / 1.30
	CDSTraj [47]	0.36 / - / -	0.86 / - / -	1.36 / - / -	2.02 / - / -	2.85 / - / -	1.49 / - / -
	GATran*	0.45 / 0.15 / 0.35	0.75 / 0.29 / 0.60	0.82 / 0.42 / 0.65	0.85 / 0.46 / 0.70	1.55 / 0.58 / 1.30	0.88 / 0.38 / 0.72
	GATran [†]	0.38 / 0.11 / 0.26	0.66 / 0.24 / 0.49	0.70 / 0.35 / 0.54	0.73 / 0.39 / 0.58	1.35 / 0.46 / 1.08	0.76 / 0.31 / 0.59
	GATran	0.34 / 0.09 / 0.22	0.61 / 0.21 / 0.44	0.65 / 0.31 / 0.48	0.68 / 0.35 / 0.51	1.27 / 0.41 / 0.97	0.71 / 0.27 / 0.52
HighD	S-LSTM [3]	0.22 / 0.23 / 0.37	0.62 / 0.41 / 0.76	1.27 / 0.60 / 1.20	2.15 / 0.81 / 1.68	3.41 / 1.03 / 2.22	1.53 / 0.62 / 1.25
	CS-LSTM [4]	0.22 / 0.21 / 0.35	0.61 / 0.39 / 0.72	1.24 / 0.57 / 1.14	2.10 / 0.77 / 1.61	3.27 / 0.99 / 2.14	1.49 / 0.59 / 1.19
	S-GAN [5]	0.30 / - / -	0.78 / - / -	1.46 / - / -	2.34 / - / -	3.41 / - / -	1.66 / - / -
	WSiP [40]	0.20 / - / -	0.60 / - / -	1.21 / - / -	2.07 / - / -	3.14 / - / -	1.44 / - / -
	MHA-LSTM [41]	0.19 / - / -	0.55 / - / -	1.10 / - / -	1.84 / - / -	2.78 / - / -	1.29 / - / -
	HMNet [44]	- / 0.20 / 0.33	- / 0.37 / 0.69	- / 0.55 / 1.09	- / 0.74 / 1.53	- / 0.94 / 2.02	- / 0.56 / 1.13
	STDAN [45]	0.19 / 0.05 / 0.07	0.27 / 0.07 / 0.13	0.48 / 0.11 / 0.28	0.91 / 0.18 / 0.59	1.66 / 0.29 / 1.08	0.70 / 0.14 / 0.43
	GaVa [48]	0.17 / - / -	0.24 / - / -	0.42 / - / -	0.86 / - / -	1.31 / - / -	0.60 / - / -
	CDSTraj [47]	0.13 / - / -	0.21 / - / -	0.32 / - / -	0.38 / - / -	1.05 / - / -	0.42 / - / -
	GATran (Ours)*	0.18 / 0.08 / 0.12	0.29 / 0.12 / 0.21	0.36 / 0.16 / 0.29	0.52 / 0.21 / 0.40	0.95 / 0.28 / 0.72	0.46 / 0.17 / 0.35
	GATran (Ours) [†]	0.14 / 0.06 / 0.09	0.24 / 0.09 / 0.16	0.30 / 0.12 / 0.22	0.42 / 0.16 / 0.31	0.78 / 0.21 / 0.55	0.38 / 0.13 / 0.27
	GATran(our)	0.12 / 0.05 / 0.07	0.20 / 0.07 / 0.13	0.26 / 0.10 / 0.18	0.37 / 0.13 / 0.25	0.68 / 0.17 / 0.45	0.33 / 0.10 / 0.22

Note: All results are reported under the setting of $k = 9$ using $\min\text{RMSE}_k(m)$, $\min\text{ADE}_k(m)$, and $\min\text{FDE}_k(m)$. Entries annotated with * and [†] denote evaluations with $k = 3$ and $k = 6$, respectively. The best results are highlighted in bold.

- 6) *MHA-LSTM* [41]: This model uses a multihead attention combined with LSTM to capture the spatiotemporal interactions between agents by considering features such as speed and vehicle type to predict trajectories.
- 7) *FHIF* [42]: This model uses multihead attention to extract historical interaction information between the intelligent agent and surrounding vehicles, and predicts future trajectories through multimodal recognition and trajectory generation modules.
- 8) *DACR-AMTP* [43]: This model analyses vehicle interactions through dynamic drivable area analysis, utilises collision risk probability to optimize direction prediction, and combines multihead attention mechanisms to capture long-term spatio-temporal dependencies.
- 9) *HMNet* [44]: This model is a context-independent hierarchical motion encoder-decoder network for vehicle trajectory prediction, which achieves trajectory prediction by extracting motion patterns of position, velocity, and acceleration through hierarchical extraction.
- 10) *STDAN* [45]: This model captures the low-level motion state of vehicles, high-level social interactions, and their dynamic temporal dependencies through a hierarchical structure. It introduces a driving intention-specific feature fusion mechanism to adaptively integrate spatio-temporal features to achieve multimodal trajectory prediction.
- 11) *iNATran* [46]: This model effectively captures social interaction between vehicles and temporal dependencies between consecutive frames by combining graph attention mechanisms and Transformer encoders, and uses stacked attention modules to extract spatio-temporal features in layers.
- 12) *CDSTraj* [47]: This model simulates traffic scenarios with intrinsic uncertainty by introducing a feature diffusion module and integrates complex contextual scene features to improve prediction accuracy.
- 13) *GaVa* [48]: This model dynamically allocates attention by introducing an adaptive visual sector mechanism and utilizes CNN and GAT to construct a dynamic traffic map to capture the spatio-temporal dependencies.

E. Quantitative Results

Table I presents the quantitative comparison between the proposed model and state-of-the-art methods on the NGSIM and HighD datasets. All methods are evaluated under the same multimodal evaluation protocol following CS-LSTM [4] for fair comparison. The results demonstrate that the proposed model consistently outperforms existing methods on both datasets under the same evaluation protocol. All reported improvements are obtained with $k = 9$. Specifically, compared to the current best-performing method, CDSTraj, the proposed model

TABLE II
ABLATION STUDY OF MODEL MODULES ON THE NGSIM AND HIGHD DATASETS

Metric	Model Variant	Prediction Horizon (s)										Ave
		1		2		3		4		5		
minRMSE	w/o Feature Fusion	0.35 ±0.01	/ 0.12 ±0.01	0.63 ±0.01	/ 0.22 ±0.01	0.68 ±0.02	/ 0.29 ±0.01	0.72 ±0.02	/ 0.41 ±0.02	1.32 ±0.04	/ 0.73 ±0.03	0.74 / 0.35
	w/o Spatio-Temporal	0.36 ±0.01	/ 0.16 ±0.01	0.66 ±0.02	/ 0.32 ±0.02	0.79 ±0.03	/ 0.42 ±0.02	1.02 ±0.04	/ 0.54 ±0.03	1.73 ±0.06	/ 0.81 ±0.04	0.91 / 0.45
	w/o MMoE	0.36 ±0.01	/ 0.13 ±0.01	0.64 ±0.01	/ 0.23 ±0.01	0.69 ±0.02	/ 0.30 ±0.01	0.73 ±0.02	/ 0.42 ±0.02	1.30 ±0.03	/ 0.75 ±0.03	0.74 / 0.37
	GATran	0.34 ±0.01	/ 0.11 ±0.01	0.61 ±0.01	/ 0.20 ±0.01	0.65 ±0.01	/ 0.26 ±0.01	0.68 ±0.01	/ 0.37 ±0.01	1.27 ±0.02	/ 0.68 ±0.02	0.71 / 0.32
minADE	w/o Feature Fusion	0.10 ±0.01	/ 0.06 ±0.01	0.23 ±0.01	/ 0.09 ±0.01	0.35 ±0.01	/ 0.13 ±0.01	0.39 ±0.01	/ 0.16 ±0.01	0.46 ±0.02	/ 0.20 ±0.01	0.31 / 0.13
	w/o Spatio-Temporal	0.11 ±0.01	/ 0.07 ±0.01	0.26 ±0.01	/ 0.12 ±0.01	0.36 ±0.02	/ 0.17 ±0.01	0.44 ±0.03	/ 0.21 ±0.02	0.54 ±0.04	/ 0.25 ±0.02	0.34 / 0.16
	w/o MMoE	0.11 ±0.01	/ 0.06 ±0.01	0.24 ±0.01	/ 0.10 ±0.01	0.36 ±0.01	/ 0.14 ±0.01	0.40 ±0.01	/ 0.18 ±0.01	0.47 ±0.02	/ 0.22 ±0.01	0.32 / 0.14
	GATran	0.09 ±0.01	/ 0.05 ±0.01	0.21 ±0.01	/ 0.07 ±0.01	0.31 ±0.01	/ 0.10 ±0.01	0.35 ±0.01	/ 0.13 ±0.01	0.41 ±0.01	/ 0.17 ±0.01	0.27 / 0.10
minFDE	w/o Feature Fusion	0.23 ±0.01	/ 0.10 ±0.01	0.46 ±0.01	/ 0.16 ±0.01	0.51 ±0.02	/ 0.21 ±0.01	0.55 ±0.02	/ 0.30 ±0.02	1.02 ±0.04	/ 0.52 ±0.03	0.55 / 0.26
	w/o Spatio-Temporal	0.25 ±0.01	/ 0.12 ±0.01	0.52 ±0.03	/ 0.22 ±0.01	0.64 ±0.04	/ 0.32 ±0.02	0.78 ±0.06	/ 0.45 ±0.03	1.35 ±0.09	/ 0.68 ±0.05	0.71 / 0.36
	w/o MMoE	0.23 ±0.01	/ 0.11 ±0.01	0.47 ±0.01	/ 0.18 ±0.01	0.52 ±0.02	/ 0.24 ±0.01	0.56 ±0.02	/ 0.33 ±0.02	1.03 ±0.03	/ 0.55 ±0.03	0.56 / 0.28
	GATran	0.22 ±0.01	/ 0.07 ±0.01	0.44 ±0.01	/ 0.13 ±0.01	0.48 ±0.01	/ 0.18 ±0.01	0.51 ±0.02	/ 0.25 ±0.01	0.97 ±0.03	/ 0.45 ±0.02	0.52 / 0.22

Note: Results are reported in the format *NGSIM/HighD* using $\min\text{RMSE}_k$, $\min\text{ADE}_k$, and $\min\text{FDE}_k$ with $k = 9$, presented as mean $\pm$ standard deviation. Bold indicate the best results.

achieves a 51% reduction in minRMSE on the NGSIM dataset and a 21% reduction on the HighD dataset. Moreover, even when evaluated with smaller values of k (i.e., $k = 3$ or $k = 6$), the proposed model still achieves highly competitive performance. Notably, the proposed model exhibits superior performance in long-term prediction horizons (4–5 s), achieving an average improvement of 61% on the NGSIM dataset and 19% on the HighD dataset in minRMSE. In comparison, for short-term prediction horizons (1–3 s), the model achieves an average improvement of 29% on the NGSIM dataset and 10% on the HighD dataset relative to CDSTraj. Experimental results show that our model achieves superior trajectory prediction, especially for long-term forecasts. This can be mainly attributed to effective feature fusion and spatio-temporal modeling, which respectively enhance input representation and scene understanding. Moreover, the decoupled multitask learning framework further improves model generalization and predictive accuracy.

F. Ablation Study

To evaluate the contribution of each major component, we conduct ablation studies on the NGSIM and HighD datasets by individually removing the feature fusion module, spatio-temporal interaction module, and MMoE module. The results are reported in Tables II and III.

1) *Feature Fusion*: Removing the feature fusion module results in a consistent performance degradation across both datasets and all evaluation metrics. On the NGSIM dataset, the absence of feature fusion leads to noticeable increases in minRMSE, minADE, and minFDE, particularly at longer prediction horizons. Similar trends are observed on the HighD dataset, though with relatively smaller margins. These results indicate that jointly encoding semantic and dynamic information is important for learning a comprehensive representation of the target vehicle, which in turn supports more stable and accurate trajectory prediction.

2) *Spatio-Temporal Interaction*: The spatio-temporal interaction module has the most significant impact on overall

TABLE III
ABLATION STUDY OF DRIVING INTENTION PREDICTION ACCURACY (MEAN $\pm$ STD)

Dataset	Model Variant	Accuracy	
		Lateral	Longitudinal
NGSIM	w/o Feature Fusion	0.97 ±0.01	0.92 ±0.01
	w/o Spatio-Temporal	0.95 ±0.02	0.90 ±0.02
	w/o MMoE	0.96 ±0.01	0.92 ±0.02
	GATran	0.98 ±0.01	0.93 ±0.01
HighD	w/o Feature Fusion	0.98 ±0.01	0.98 ±0.01
	w/o Spatio-Temporal	0.97 ±0.01	0.96 ±0.01
	w/o MMoE	0.97 ±0.01	0.97 ±0.02
	GATran	0.99 ±0.01	0.99 ±0.01

Note: Bold indicate the best results.

performance. On the NGSIM dataset, removing this module results in substantial performance degradation, increasing minRMSE, minADE, and minFDE by approximately 22%, 21%, and 27%, respectively. This highlights the critical role of explicit spatio-temporal interaction modeling in dense traffic scenarios, where vehicle behavior is strongly influenced by surrounding agents. On the HighD dataset, the removal of the spatio-temporal interaction module also leads to pronounced performance drops, especially for long-term predictions. This is because the spatio-temporal interaction module integrates spatial interaction modeling with temporal encoding, enabling the model to capture how interagent influences evolve over time. Without this module, the model struggles to effectively reason about dynamic interactions, resulting in larger long-horizon prediction errors.

3) *MMoE*: The impact of the MMoE module differs across datasets. On the NGSIM dataset, removing MMoE causes a moderate but consistent increase in prediction errors, suggesting that shared representations alone can already capture a large portion of the motion patterns in relatively homogeneous traffic scenarios. In contrast, on the HighD dataset, the MMoE module plays a more prominent role. Its removal leads to average performance degradations of approximately 13% in minRMSE,

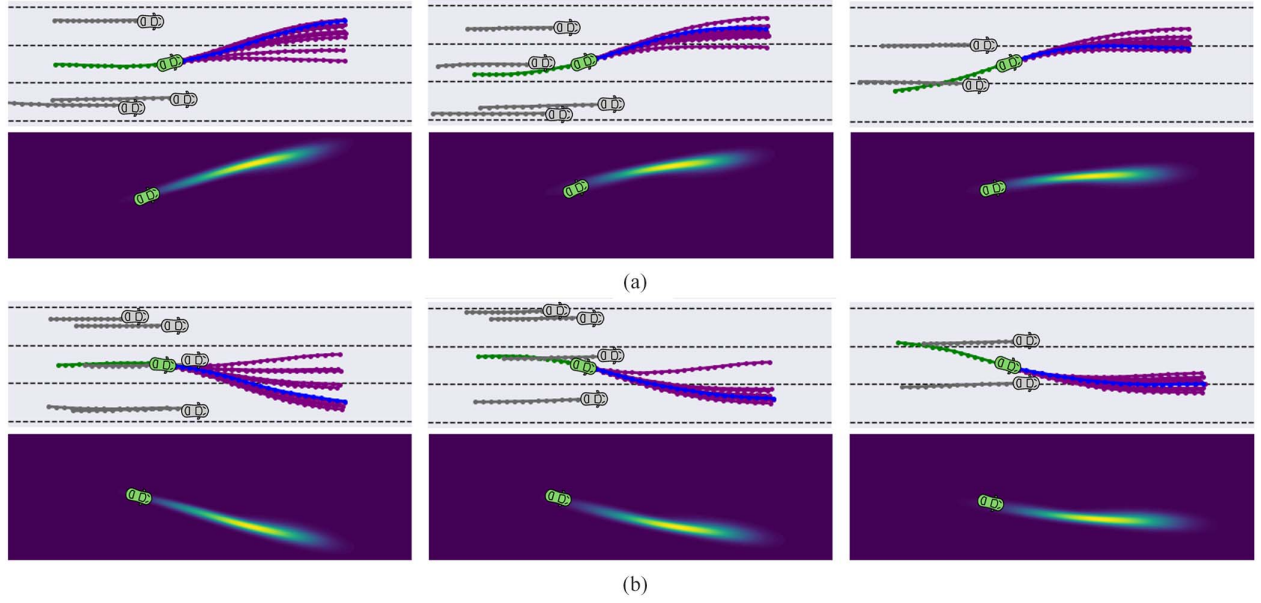


Fig. 5. Multimodal trajectory prediction visualization. The **blue line** represents the ground truth trajectory of the target vehicle. The **purple line** represents the multimodal trajectory prediction of the target vehicle. The **green line** and gray line represent the historical trajectory of the target vehicle and the surrounding vehicles, respectively. The **green car** and gray car represent the target vehicle and the surrounding vehicles, respectively.

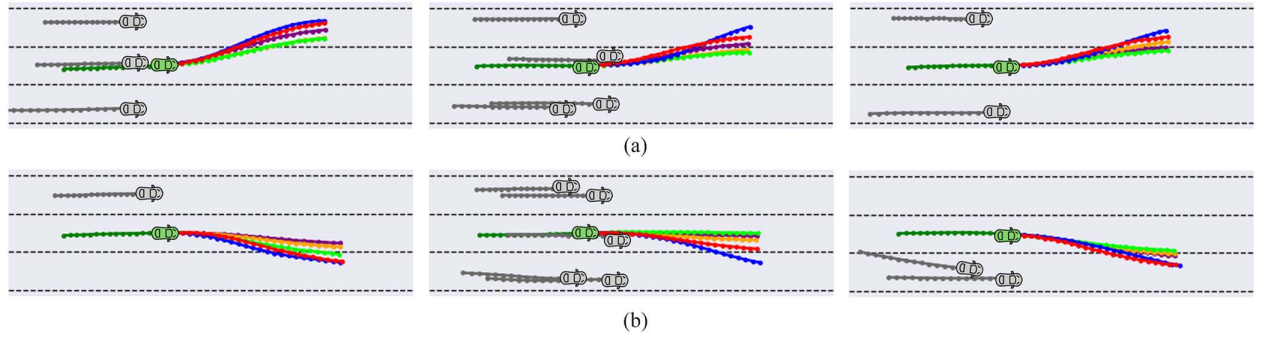


Fig. 6. Trajectory prediction visualization of ablation study. The **blue line** represents the ground truth trajectory of the target vehicle. The **red line** represents the predicted trajectory of GATran, the **purple line** represents the predicted trajectory of the ablation model without feature fusion module, the **lime line** represents the predicted trajectory of the ablation model without spatio-temporal interaction module, and the **orange line** represents the predicted trajectory of the ablation model without MMOE module. The **green line** and gray line represent the historical trajectory of the target vehicle and the surrounding vehicles, respectively. The **green car** and gray car represent the target vehicle and the surrounding vehicles, respectively.

23% in minADE, and 15% in minFDE. This can be attributed to the greater diversity of driving behaviors in HighD, where the MMoE design effectively facilitates task-specific feature disentanglement within the multitask learning framework, thereby improving both trajectory prediction and intention recognition.

4) *Intention Prediction*: Overall, the full model consistently achieves the highest accuracy for both lateral and longitudinal intention prediction, demonstrating the effectiveness of the proposed multitask learning framework. On the NGSIM dataset, removing any of the three modules leads to a noticeable decrease in prediction accuracy. In particular, excluding the spatio-temporal interaction module results in the most significant performance degradation for both lateral and longitudinal intentions. This suggests that explicit modeling of spatio-temporal interactions is crucial for accurate intention

inference in dense traffic environments, where driver decisions are strongly influenced by surrounding vehicles. The feature fusion module also contributes to performance improvement by enabling more effective integration of semantic and dynamic information, while the MMoE module provides additional gains by facilitating task-specific representation learning. On the HighD dataset, all model variants achieve relatively high accuracy, reflecting the structured highway scenarios and cleaner motion patterns captured by drone-based data. Nevertheless, the full model still consistently outperforms the ablated variants, particularly for longitudinal intention prediction. The performance gain from the MMoE module is more evident on HighD, indicating its effectiveness in handling the greater diversity of driving behaviors present in this dataset through improved task disentanglement.

G. Visualization

1) *Multimodal Trajectory Prediction*: To evaluate the prediction performance of the proposed model, we present a visualization of the multimodal trajectory prediction for the target vehicle in Fig. 5. Two challenging traffic scenarios are selected from the dataset, with Fig. 5(a) and (b) illustrating the left and right scenarios, respectively. The figure is organized into two rows: the first row depicts the visualization of multimodal trajectory predictions across continuous time steps, while the second row presents a heatmap representing driving intention predictions. From the visualization, it can be observed that at the onset of lane-changing scenarios, the target vehicle exhibits more discrete multimodal trajectory predictions. As the lane change progresses, these predictions become increasingly convergent. Furthermore, the heatmap provides a more intuitive representation of the evolution of driving intention predictions, which aligns with the trends observed in the multimodal trajectory predictions.

2) *Ablation Study*: To further evaluate the effectiveness of each module in the proposed model, we visualize the results of the ablation study in Fig. 6. Several representative traffic scenarios are selected from the dataset to illustrate the prediction performance of the proposed model. The visualization reveals that the proposed model achieves superior performance, particularly in complex interaction scenarios. For instance, in the right lane-change scenario shown in the second panel of Fig. 6(b), where multiple surrounding vehicles are present near the target vehicle, the ablation models tend to adopt more conservative, straightforward prediction strategies. In contrast, the proposed model provides more accurate and consistent trajectory predictions. This improved performance can be attributed to the model's ability to capture latent motion patterns, such as the observation that vehicles on the left lane tend to move slower than those on the right, thereby creating a gap that facilitates the target vehicle's lane change.

V. CONCLUSION

In this article, we propose a novel model named GATran for multimodal trajectory prediction and driving intention estimation. Different from existing methods, the article decouple the motion prediction of the target vehicle into two subtasks, trajectory prediction and driving intention estimation, thus model the motion prediction as a multitask learning problem. Additionally, the model introduces the spatial interaction module to capture the spatio-temporal dependencies between the target vehicle and surrounding vehicles, and the effective feature fusion module to fuse the historical trajectory with abundant contextual information to overcome the limitations of data sparsity and complex spatio-temporal interaction. The experimental results on the NGSIM and HighD datasets demonstrate the effectiveness of the proposed model, which outperforms the state-of-the-art methods by a large margin. The ablation study further verifies the effectiveness of each module in the proposed model. The model displays the ability to capture the latent motion patterns and complex spatio-temporal interactions, which is crucial for multimodal trajectory prediction and driving intention

estimation in complex traffic scenarios. However, individual motion prediction not be sufficient for real-world applications, a reliable autonomous driving system should consider the motion prediction and downstream tasks, such as motion planning and control, which will be our future work.

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